

Overview

Build an **AI-powered Fraud & Risk Detection Platform** that analyzes business transactions and customer activities to detect suspicious or unusual behavior.

The system should not simply show transaction records. It should use AI and rule-based analysis to calculate a **Risk Score**, identify suspicious patterns, explain why an activity is risky, and alert the business/admin.

The goal is to build something that could be used by **e-commerce companies, fintech businesses, online marketplaces, subscription platforms, and payment-related businesses.**

1. Authentication & Users

Create secure authentication for:

- Admin
- Business Manager
- Analyst

Each role should have different permissions.

2. Transaction Management

The system should allow businesses to:

- Add transactions manually
- Import transactions using CSV
- Receive transactions through API
- View all transactions
- Search and filter transactions
- View complete transaction details

Transaction data can include:

- Customer ID
- Transaction ID
- Amount

- Date/time
- Payment method
- IP address
- Location
- Device information
- Customer account age
- Previous transaction history

3. AI Risk Scoring

Every transaction should automatically receive a **Risk Score from 0–100**.

Example:

- 0–30 → Low Risk
- 31–70 → Medium Risk
- 71–100 → High Risk

The AI should consider multiple factors instead of only transaction amount.

For example:

- Unusual transaction amount
- New device
- New location
- Multiple transactions in a short period
- Multiple accounts using the same device
- Multiple customers using the same IP
- Sudden change in customer behavior
- Unusual login/activity pattern
- High-value transaction from a new account

4. AI Explanation

The system must explain **why** a transaction received its risk score.

Example:

Risk Score: 87

High Risk because:

- Customer normally spends \$50–\$100.
- Current transaction is \$1,200.
- New device detected.
- Login location is different from previous activity.
- Three transactions occurred within 5 minutes.

The explanation should be understandable to a normal business user.

5. Fraud Pattern Detection

The system should detect suspicious patterns across multiple transactions.

Examples:

Rapid Transactions

Detect many transactions from the same customer within a short time.

Device Sharing

Detect multiple customer accounts using the same device.

IP Sharing

Detect suspicious numbers of accounts using one IP address.

Location Anomaly

Detect transactions from unusual or impossible locations.

Account Behavior Change

Compare current customer behavior with historical behavior.

Transaction Amount Anomaly

Identify transactions that are significantly different from the customer's normal spending pattern.

6. AI Anomaly Detection

Implement an AI/ML-based anomaly detection component.

The system should learn from historical transaction data and identify transactions that do not match normal behavior.

Interns should experiment with suitable ML techniques such as:

- Isolation Forest
- Clustering
- Statistical anomaly detection
- Classification models

The model should produce an anomaly score that can contribute to the final risk score.

7. Rules Engine

Create a configurable rules system.

Admin should be able to create rules such as:

- If transaction > \$5,000 → increase risk
- If more than 5 transactions in 10 minutes → suspicious
- If new device + high-value transaction → high risk
- If multiple accounts use same IP → investigate

Rules should be manageable from the dashboard.

8. Risk Decision Engine

Combine:

AI/ML Score + Rules + Customer History + Transaction Behavior

to generate the final risk score.

Example:

ML Anomaly Score	72
Rule Score	85
Customer Behavior	90

Final Risk Score	87

The exact scoring architecture should be designed by the intern.

9. Fraud Alerts

When a high-risk activity is detected:

- Create an alert
- Show alert on dashboard
- Store the reason
- Assign severity
- Notify the relevant user

Alert statuses:

- New
- Investigating
- Confirmed Fraud
- False Positive
- Resolved

10. Investigation System

An analyst should be able to open a suspicious transaction and investigate it.

Show:

- Customer information
- Transaction history

- Related transactions
- Devices
- IP addresses
- Locations
- Risk factors
- AI explanation
- Related alerts

The analyst should be able to add investigation notes.

11. Customer Risk Profile

Every customer should have a risk profile.

Example:

Customer: CUST-1029

Risk Level: Medium

Risk Score: 64

Total Transactions: 128

Suspicious Transactions: 7

Devices Used: 4

Locations Used: 3

Previous Fraud Reports: 1

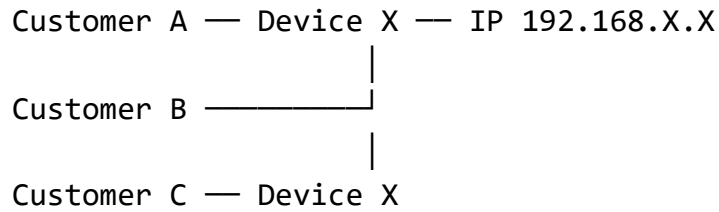
The system should continuously update the profile.

12. Fraud Network Detection

Build a visual relationship view showing connections between:

Customer → Device → IP → Transaction → Location

Example:



This can help analysts discover groups of related suspicious accounts.

13. Dashboard

Create an admin dashboard showing:

- Total transactions
- High-risk transactions
- Medium-risk transactions
- Fraud alerts
- Confirmed fraud
- False positives
- Average risk score
- Fraud trends
- Suspicious customers
- Suspicious devices/IPs

Include charts and filters.

14. AI Investigation Assistant

Add an AI assistant specifically for fraud analysts.

The analyst can ask:

"Why is this customer suspicious?"

"Show me unusual activity from this customer."

"What transactions are connected to this device?"

"Summarize this investigation."

The AI should answer using the platform's actual transaction and investigation data.

15. Feedback & Model Improvement

When an analyst marks an alert as:

- Confirmed Fraud
- False Positive

store that feedback.

Use this data as future training/evaluation data so the fraud detection model can be improved over time.

16. Reports

Generate reports for:

- Daily fraud activity
- Monthly fraud activity
- High-risk customers
- High-risk transactions
- Confirmed fraud
- False positives
- Fraud trends

Allow reports to be exported.

17. API

Create APIs for external businesses to send transactions to the system.

Example:

POST /api/transactions

POST /api/risk-check

GET /api/transactions/{id}

GET /api/risk/{transaction_id}

POST /api/alerts/{id}/review

An external e-commerce/payment system should be able to send a transaction and receive:

Risk Score: 87

Risk Level: HIGH

Decision: REVIEW

18. Real-Time Detection

The system should support real-time transaction analysis.

Flow:

New Transaction



Data Validation



Rule Engine



ML/AI Analysis



Customer History



Risk Engine



Risk Score



Decision



Alert / Approve / Review

19. Technology Requirements

Frontend

- React / Next.js
- Dashboard
- Charts
- Investigation interface
- Fraud network visualization

Backend

- Python
- FastAPI
- PostgreSQL

AI/ML

- Python ML libraries
- Anomaly detection
- Classification where appropriate
- LLM for explanations and investigation assistant

Background Processing

- Redis
- Celery or equivalent queue system

Security

- JWT authentication
- Role-based access
- API authentication
- Audit logs
- Secure handling of sensitive transaction data

Final Deliverable

The intern must deliver a **working AI Fraud & Risk Detection product**, not just an ML model.

It must include:

- Authentication
- Transaction management
- CSV import
- Transaction API
- AI/ML anomaly detection
- Risk scoring
- Rules engine
- AI explanations
- Fraud alerts
- Customer risk profiles
- Investigation system
- Fraud network visualization
- AI investigation assistant
- Dashboard
- Reports
- Feedback system
- Database
- APIs
- Deployment